# Week 1 reading guide and worked examples

## Weekly overview

### Learning outcomes

By the end of this week, you should be able to:

- interpret and use set, function, quantifier, implication and equivalence notation, and complete short arguments by direct reasoning, contradiction or induction (`M1`);
- manipulate powers, logarithms, functions, indexed sums, products and finite or convergent geometric series (`M2`);
- calculate and interpret derivatives and elementary integrals, construct first- and second-order Taylor approximations, and solve elementary one-variable optimisation problems (`M3`); and
- represent probability questions as events and apply set operations, probability rules and elementary counting (`S1`).

### Prerequisites and retrieval

Retrieve fraction arithmetic, exponent rules, elementary equation solving and the meaning of a function before beginning the readings. No earlier course outcome is a formal prerequisite for Week 1. Write down any rule that you cannot reproduce reliably and revisit it during the first study block.

### Student materials

Use the Week 1 reading guide for exact written sections and skip instructions. Then study and reproduce the original worked examples. Complete the exercise sheet, consulting progressive hints only after a serious attempt. Use the complete solutions to mark your work after the attempt is finished. End with the 25-minute self-test and update your error log from its solutions.

The external readings remain on their authors' or institutions' sites. Open only the assigned sections. No video, applet, code or specialist software is required.

Before beginning the Week 1 readings, use the required notation and conventions bridge note. It is point-of-need support within the existing reading and worked-example allocation, not additional workload.

### Suggested study sequence

1. Retrieve fraction and exponent rules and write the four outcome IDs at the top of your notes.
2. Complete the readings in the listed order. Close each source and reproduce at least one important definition, argument or worked step for each outcome.
3. Study the four institutional worked examples. Cover each solution, follow the method and reproduce the work from a blank page.
4. Attempt all questions without hints or solutions. A basic scientific calculator is permitted only for the logarithmic estimate in `W01-Q03-PB`; it is not required elsewhere.
5. If you are stuck, use the hints in order and stop as soon as you have a useful next step.
6. Mark from the full solutions. Record the first cause of each error under `M1`, `M2`, `M3` or `S1`.
7. Take the 25-minute self-test without notes, mark it and add any new cause of error to the same outcome-labelled log.
8. Reattempt each incorrect or incomplete part from a blank page two or three days later.

### Workload and required outputs

The times below are provisional planning guidance, not a guarantee, and have not yet been verified by students. They are subject to first-cohort calibration. Students in the first live course run will be invited to report their actual component times so the plan can be recalibrated for the following year.

| Component | Target time | Required output |
|---|---|---|
| Required external reading, including brief closed-source summaries | 3 h 40 min | Brief original notes and at least one reproduced definition, argument or worked step for each outcome |
| Institutional worked-example study and reproduction | 1 h 5 min | Four examples reproduced from a blank page |
| Week 1 problem sheet | 4 h 15 min | All 17 scorable parts attempted and marked |
| Self-test | 25 min | All four self-test parts attempted and marked |
| Marking, error log and first spaced reattempt | 35 min | One error-log entry for each outcome, including "no error yet" where applicable |
| **Core total** | **10 h** | Completed Week 1 study cycle |

The problem sheet contains 10 numbered questions, 17 scorable parts, 100 marks and a 255-minute target. Its mark balance is 30% foundation, 45% standard multi-step work, 15% explanation or interpretation, and 10% stretch or synthesis. The reading and worked-example component totals 4 hours 45 minutes. The self-test, marking, error logging and first reattempt total one hour.

The self-test retains four scorable parts, 20 marks and a 25-minute target. The separate equivalence and indexed-product checks are embedded in existing Week 1 exercise parts by replacing redundant marked work; the weekly total does not increase.

### End-of-week check

Before moving on, make sure that you can explain why a condition or endpoint check is needed rather than merely reproduce a calculation. Keep the item ID beside every error-log entry so that you can reattempt the exact part later.

## Required reading route

### How to use this route

Complete the six required written selections in order. The locators and skip instructions below implement the approved Week 1 syllabus. They do not assign the surrounding books, courses, exercises or media. Spend about 220 minutes on the assigned reading, including the brief closed-source summaries and reproduction prompts below. The separate 65-minute allocation in `worked_examples.md` is for the four institutional examples and is not part of this reading allocation.

### Required bridge note

Before the external selections, read the required notation and conventions bridge note. It standardises course notation and is point-of-need support within the existing Week 1 reading and worked-example allocation, not additional workload.

### 1. Logic, functions and induction (`M1`)

**Resource `MATH-M1-OSB-REQ`; source `OSBORNE_MM`.** In Martin J. Osborne, *Mathematical Methods for Economic Theory*, read Section 1.1, "Logic: basics and proof by induction" and only "Intervals" and "Functions" in Section 1.4.

**Skip:** "Continuous functions" and all later distance and epsilon–delta material in Section 1.4. Do not use interactive or graphical elements.

Target reading time: 35 minutes.

### 2. Sets and elementary proof methods (`M1`)

**Resource `MATH-M1-HEF-REQ`; source `HEFFERON_LA`.** In Jim Hefferon, *Linear Algebra*, use the book PDF to read Appendix A-1 "Statements", A-2 "Quantifiers", A-3 to A-4 "Techniques of Proof", and A-5 to A-12 "Sets, Functions, and Relations".

**Skip:** all linear-algebra chapters for this assignment. Stop after "Sets, Functions, and Relations".

Target reading time: 20 minutes.

### 3. Logarithms, sequences and geometric series (`M2`)

**Resource `MATH-M2-OPS-REQ`; source `OPENSTAX_PRECALC2E`.** In Jay Abramson, *Precalculus 2e*, read the assigned parts of:

- Section 4.3, "Logarithmic Functions": the definition, exponential–log conversions and natural logarithms;
- Section 4.5, "Logarithmic Properties": the product, quotient and power rules;
- Section 11.1, "Sequences and Their Notations": explicit notation only;
- Section 11.3, "Geometric Sequences"; and
- Section 11.4, "Series and Their Notations" through infinite geometric series.

**Skip:** arithmetic sequences and arithmetic-series formulae, annuities, graphing, calculator and technology exercises, all Media links, online practice tools, and applications not needed for the stated algebraic skill.

Target reading time: 45 minutes.

### 4. One-variable calculus and optimisation (`M3`)

**Resource `MATH-M3-OSB-REQ`; source `OSBORNE_MM`.** In Osborne, *Mathematical Methods for Economic Theory*, read Section 1.5, "Calculus: one variable", the "One variable" part of Section 5.1 through the interval-maximisation procedure, and the "One variable" part only of Section 5.2.

**Skip:** every "Many variables" subsection in Sections 5.1 and 5.2, rotatable graphs, and proofs beyond the stated operational first- and second-order conditions.

Target reading time: 40 minutes.

### 5. Taylor approximation (`M3`)

**Resource `MATH-M3-OPS-REQ`; source `OPENSTAX_CALC2`.** In Gilbert Strang and Edwin "Jed" Herman, *Calculus Volume 2*, read only "Taylor Polynomials" through the definition, Example 6.11 and Checkpoint 6.10, then Example 6.13 and Checkpoint 6.12 in Section 6.3, "Taylor and Maclaurin Series".

**Skip:** power-series prerequisites and convergence, proofs of Taylor's theorem, general remainder estimates, all `[T]` technology items, graphs, the student project, Taylor-series manipulations and every exercise after 123.

Target reading time: 20 minutes.

### 6. Probability events, rules and counting (`S1`)

**Resource `STAT-S1-REQ-MIT`; source `MIT_1805`.** On Jeremy Orloff and Jonathan Bloom's MIT OpenCourseWare 18.05 classes and written-materials page, read Class 1 Reading 1b Sections 2.2–2.4.4 and Class 2 Reading Sections 2–3.

**Skip:** Problem Set 1 Problem 2, which uses a video hint; Problem 3(e, g), which uses R; and all linked videos, interactive questions and applets.

Target reading time: 60 minutes.

**Reading output**

After each selection, close the source and write a brief original summary. Across the six selections, reproduce at least one example or argument for each of `M1`, `M2`, `M3` and `S1`. Do not copy external explanations or exercise prompts into shared notes.

## Worked examples

These original institutional examples support `M1`, `M2`, `M3` and `S1`. For each example, cover the **Solution**, follow the **Method**, and reproduce the work from a blank page before reading the **Interpretation**. Allow about 65 minutes for study and reproduction.

**Example 1 — a direct argument (`M1`)**

Show that the square of every even integer is even.

**Method.** Begin with the definition of an even integer and use direct reasoning. The conclusion must have the same definition.

**Solution.** Let $n \in \mathbb{Z}$ be even. Then there is an integer $k$ such that $n = 2k$. Therefore,

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

Because $2k^2 \in \mathbb{Z}$, the number $n^2$ is twice an integer and is therefore even.

**Interpretation.** The argument does not rely on a few numerical examples. It begins with an arbitrary even integer, so it establishes the implication for every even integer.

**Example 2 — a declining geometric stream (`M2`)**

A programme pays 80 units at $t = 0$, and each later payment is one half of the preceding payment. Find the infinite total. Then find the smallest final included period $N$ for which the cumulative total over $t = 0, \dots, N$ is at least 150.

**Method.** Identify the first term and common ratio, apply both geometric sum formulae, and verify the integer boundary.

**Solution.** The common ratio is $r = 1/2$, so $|r| < 1$ and the infinite series converges. Its total is

$$\sum_{t=0}^{\infty} 80 \left(\frac{1}{2}\right)^t = \frac{80}{1 - 1/2} = 160.$$

The finite total is

$$S_N = 80 \frac{1 - (1/2)^{N+1}}{1 - 1/2} = 160 \left(1 - 2^{-(N+1)}\right).$$

The threshold condition is

$$\begin{aligned} 160 \left(1 - 2^{-(N+1)}\right) &\geq 150, \\ 2^{-(N+1)} &\leq \frac{1}{16} = 2^{-4}. \end{aligned}$$

Because $2^x$ is strictly increasing, $-(N+1) \leq -4$, so $N \geq 3$. The boundary check gives

$$S_2 = 140 < 150, \qquad S_3 = 150.$$

Thus the smallest final period is $N = 3$.

**Interpretation.** The index starts at zero, so $N = 3$ includes four payments. The boundary check prevents confusing the final index with the number of included periods.

**Example 3 — a global maximum on an interval (`M3`)**

Let

$$p(q) = 12q - q^2, \qquad q \in [0, 10].$$

Find the global maximum and the maximising value of $q$.

**Method.** Find every interior stationary point and compare its value with both endpoints. Use curvature as an additional sufficiency check.

**Solution.** The first two derivatives are

$$p'(q) = 12 - 2q, \qquad p''(q) = -2.$$

The unique stationary point is $q = 6$, which lies inside the feasible interval. The candidate values are

$$p(0) = 0, \qquad p(6) = 36, \qquad p(10) = 20.$$

Therefore the global maximum is 36, attained at $q = 6$. The strict concavity $p''(q) < 0$ confirms that there is at most one maximiser.

**Interpretation.** A first-order condition identifies a candidate. The endpoint comparison is what establishes the global conclusion on the stated closed interval.

**Example 4 — counting with a complement (`S1`)**

Two representatives are selected uniformly from five people, and order does not matter. Two of the five are designated priority representatives. Find the probability that at least one priority representative is selected.

**Method.** Count all unordered pairs and subtract the pairs that contain no priority representative. Reconcile this with a direct count.

**Solution.** There are

$$\binom{5}{2} = 10$$

equally likely pairs. A pair with no priority representative must choose both people from the other three, giving $\binom{3}{2} = 3$ pairs. Hence

$$\mathbb{P}(\text{at least one priority representative}) = 1 - \frac{\binom{3}{2}}{\binom{5}{2}} = \frac{7}{10}.$$

Directly, there are $2 \cdot 3 = 6$ pairs with exactly one priority representative and one pair with both, for the same favourable total of seven.

**Interpretation.** The events “exactly one” and “both” are disjoint, so their counts may be added without double-counting.

**Reproduction check**

Before starting the exercise sheet, reproduce the defining step and final check from each example. If you cannot explain why the condition $|r| < 1$, the endpoint comparison or the complement count is needed, return to the relevant example rather than memorising only its final answer.